Cognisom

by Eyentelligence, Inc.

A patient's own sequencing drives a mechanistic simulation of their tumour and immune system, predicting which of their mutations their own immune system can see.

Variants in, mechanism out: somatic calling on NVIDIA Parabricks, HLA typing, neural peptide–MHC binding, AlphaFold2-Multimer structure through NVIDIA BioNeMo, then a ten-module agent-based tumour–immune simulation that renders in NVIDIA Omniverse. Every parameter is traceable to a published measurement or flagged as an estimate.

cognisom.com

Eyentelligence, Inc. · Delaware C-corporation, incorporated June 2026 · Flower Mound, TX

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Harvard Innovation Labs venture

NVIDIA Inception member

Research use only — not a clinical decision tool

What we are asking for

Two things, and the second matters as much

1. Experimental evidence. Our predictions are wet-lab-testable and untested; GenScript credits would synthesise the peptides we predict and measure whether they bind.

2. Research-level scientific expertise. Built AI-first by one founder, the engineering is ahead of the domain review. We are actively seeking immunology and tumour-biology experts to validate and prioritise direction.

Where NVIDIA already sits

Five technologies, in the pipeline today

BioNeMo models · Parabricks · Omniverse · Data Center GPUs (L40S) · NVIDIA Warp. Plus Bio-USD, our draft open OpenUSD schema for biological data, published under Apache 2.0 with a citable DOI.

Choosing an immunotherapy candidate means predicting whether a patient's immune system will engage their tumour. Today's tools each answer half of that, so a failure teaches you nothing.

The gap, concretely

- **Binding predictors** return a ranked peptide list with no mechanism. Nothing tells you whether a predicted epitope matters in a tumour that is hypoxic, immune-excluded, or already selecting for escape.
- **Mechanistic systems-biology models** carry that context but are population-average, so they cannot be driven by one patient's variants and HLA type.
- Neither answers *why* a candidate should work in a specific tumour. So a negative result is uninformative and iteration is slow and expensive.

Who has the problem

- Immuno-oncology discovery and translational teams in biotech and pharma, selecting neoantigen and combination candidates
- Academic cancer-immunology labs testing mechanistic hypotheses about the microenvironment
- End user: the computational biologist or translational scientist deciding which candidates go into assays
- Application area: **preclinical immuno-oncology R&D** — neoantigen selection and combination hypothesis generation

We are not proposing a therapeutic and not proposing a clinical tool. The product is software and simulation sold into discovery, which is why the market we compete in is biosimulation rather than oncology therapeutics.

The same pipeline runs from a patient's raw variants to a simulation of their own tumour–immune system, with the patient's HLA type deciding what is visible.

Variants

VCF ingest and annotation; matched tumour–normal **somatic calling on NVIDIA Parabricks**, so germline variants are not mistaken for drivers

What is presentable

HLA typing, then neural peptide–MHC binding (MHCflurry) against **that patient's own alleles** — not a reference panel

Structure

NVIDIA BioNeMo hosted NIMs: AlphaFold2-Multimer predicts the peptide–MHC complex, so an epitope becomes a modelled structure rather than a score

Mechanism

A ten-module agent-based simulation: cancer, normal, T, NK and macrophage cells under literature-sourced rules. Exports to OpenUSD and renders in **Omniverse**

Mechanistic, not predictive

Cells follow rules drawn from published measurements and the tumour–immune outcome *emerges*. So a result decomposes into named causes a scientist can inspect and dispute — the property a drug programme needs and a black-box score cannot give.

Provenance by construction

Library parameters carry their source (PMID) or are explicitly flagged ESTIMATE. We can say which of our numbers are measured and which are guesses, per parameter. Coverage is partial and openly tracked.

Deterministic decision layer

A multi-agent tumour board ranks treatments with traceable evidence, model cards and provenance, under an FDA clinical-decision-support framing. It is **deterministic by design** — no ML model, no LLM — because auditability is the point.

The Omniverse step is a file handoff, stated precisely: the engine writes a USD stage and Isaac Sim opens it. It is not a live bidirectional coupling.

On a public reference case the pipeline narrows 1,495 variants to two somatic drivers and three predicted binders on that patient's own HLA — each one a wet-lab-testable claim.

1,495 → 2

SEQC2 HCC1395 breast cancer: 1,495 variants, 29 coding, **2 surviving somatic filtering** — ARID1A and NOTCH1

3

predicted strong binders on this patient's typed HLA alleles, via neural pMHC prediction

3,841

automated tests passing, 9 skipped, across the engine, library and genomics paths

k = 1.05

whole-engine cost is near-linear in cell count; about 2.1 s per simulated hour at 2,000 cells

Why the filtering matters

Two well-known breast-cancer genes, ESR1 and ERBB2, appear in the raw calls and **do not survive** somatic filtering on this sample. A pipeline that reported them would look more impressive and be wrong. Reporting two genes instead of four is the result.

Engineering the long runs need

- Checkpoint and restart, verified bit-identical to an uninterrupted run across 17 observables including the RNG stream position
- Scaling measured and re-measured per module, not asserted

HCC1395 is a public tumour–normal reference pair, which is the point: anyone can re-run this. No patient data of our own is used, and none appears in this deck.

A modelled mechanism was tested against human tumour data the model never saw, and the prediction survived. It is a surviving negative prediction, not a confirmation — and we say so.

The test

- **Prediction.** Our MHC-I mechanism encodes hypoxia destroying MHC-I *protein* through the unfolded protein response while HLA mRNA rises. If that is right, a hypoxia signature must **not** correlate negatively with HLA transcript abundance across human tumours.
- **Data.** 592 TCGA colorectal tumours, cBioPortal, RNA-Seq V2 — the same study, assay and statistic the source paper used.
- **Result.** With IFN-γ partialled out, no MHC-I gene is negatively correlated with hypoxia. The prediction survives.
- **What it is not.** mRNA is not surface protein; bulk expression mixes tumour, stroma and immune cells; and a null is consistent with the model rather than proof of it.

Three traps found while running it

- GAPDH was used as a negative control. It cannot be one — it is a glycolytic enzyme and a canonical HIF target — and it came out at rho +0.41 with the hypoxia score.
- On raw RSEM *every* gene correlated positively, beta-actin included. Something sample-wide inflates everything, so each value became a within-sample log ratio.
- Hypoxia and IFN-γ are themselves correlated (rho +0.33) and every gene in the panel is IFN-γ inducible. The partial correlation is what the result actually rests on.
- **Each of the three would have produced a confident wrong answer.**

We report our nulls on the same footing. A separate pre-specified outcome test on a clinical cohort came back null at p = 0.55. It is on the record and has not been retired or re-specified. A platform whose negative results are visible is the only kind whose positive results mean anything.

This is the whole of our external validation. One mechanism, one cohort, one surviving negative prediction. Everything else in the system is internally consistent and externally untested — which is precisely the gap this programme would close.

Five NVIDIA technologies are in the pipeline today, spanning sequence to structure to simulation to render — and the schema we built to carry biology in OpenUSD is already public.

BioNeMo models

Eleven hosted NIM clients. **AlphaFold2-Multimer** predicts peptide–MHC complexes inside the neoantigen path itself; ESM-2, OpenFold3, Boltz-2, DiffDock, MolMIM, GenMol, RFdiffusion, ProteinMPNN, Evo2 and MSA-Search are integrated alongside it.

Parabricks

Matched tumour–normal somatic variant calling on an L40S — the front of the pipeline, and what makes the two-gene result on HCC1395 defensible rather than a germline artefact.

Omniverse & OpenUSD

A run exports to USD and streams in Isaac Sim 6.0 at 37–50 fps, so a simulation is something a scientist watches and interrogates rather than a table they read.

Data Center GPUs

L40S for the genomics and rendering stages.

NVIDIA Warp

Warp kernels back the physics paths and initialise on every run. The agent-based engine itself is CPU and measured near-linear; we do not claim it is GPU-resident.

Bio-USD — our contribution

A draft open schema for biological entities in OpenUSD. Published Apache 2.0 with a citable concept DOI (10.5281/zenodo.23603569) and engaged directly by Pixar's team on the OpenUSD forum. Infrastructure the field can adopt, not a feature we keep.

We are an Omniverse and BioNeMo user with a published OpenUSD contribution, so the partnership conversation is about an ecosystem contribution rather than a favour.

Synthesise the peptides we predict from a patient's own variants and HLA type, and measure whether they bind. The prediction is registered before the assay runs, so either outcome is informative.

The experiment

- **Input.** The three peptides our pipeline predicts as strong binders on HCC1395's own typed HLA alleles, plus predicted non-binders from the same sample as negative controls.
- **GenScript capability.** Peptide synthesis, then binding measurement against those alleles — Experimental Validation and Closed-Loop Learning.
- **Registered in advance.** The peptide list and the predicted call for each are fixed before synthesis. No post-hoc selection is possible.

Why either outcome pays

- **If they bind:** the patient-specific neoantigen path has experimental evidence for the first time, and the claim changes from in-silico to evidenced.
- **If they do not:** the error localises to presentation scoring rather than to the simulation, which is a far more useful failure than a model that was never tested.
- **Either way:** protein expression and functional validation would then let us replace estimated immune-engagement constants with measured ones — the closed loop that turns a mechanistic model into a calibrated one.

And a second ask, which we rate as highly as the credits. Cognisom is built AI-first, which is why one founder covers genomics, simulation, parameter sourcing and rendering. The consequence is that the engineering is ahead of the domain review: our open questions are now scientific, not technical — which immune mechanisms deserve priority, which estimated constants most need measuring, which predictions are worth pre-registering next. This programme offers introductions to scientific experts, and that is the input we most need. We are looking to add research-level immunology, tumour-biology and translational expertise to validate and steer direction.

We sell software and simulation into drug discovery, so our market is biosimulation. Traction is technical, not commercial, and we would rather say so than inflate it.

The market we compete in

- **Biosimulation: USD 5.7 bn in 2026**, from 4.9 bn in 2025, forecast to 25.1 bn by 2035 at a 17.9% CAGR.
- Led by Simulations Plus, Certara, Dassault Systèmes, Schrödinger and ACD/Labs, together about 35% share — an established category with public comparables, not a category we have to invent.
- **The demand driver sits one layer out:** personalised neoantigen cancer vaccines, USD 1.4 bn in 2026, forecast to 12.8 bn by 2036 at 24.6%. That is the market our *customers* chase — not revenue available to us.
- Both are third-party vendor estimates and we treat them as directional.

Traction, stated plainly

- **No pilots, letters of intent, design partners or grants to date.** Effort has gone into making the prediction path real and falsifiable before taking on commitments we could not support.
- What exists: a closed loop on public reference data, one externally tested mechanism, a published open standard with a DOI, and five NVIDIA technologies in production use.
- The next milestone is deliberately experimental rather than commercial — the pre-registered binding assay on the previous slide.
- Institutional: Cognisom is a **Harvard Innovation Labs** venture, and Eyentelligence is an **NVIDIA Inception** member.
- **No formal scientific advisor yet** — early conversations with a Harvard faculty member, and nothing signed. Closing that is what we want from this programme's introductions.

Sources: biosimulation figures, Global Market Insights, Biosimulation Market, accessed 2 November 2026. Neoantigen cancer vaccine figures, Meticulous Research, accessed 2 November 2026.

A platform is only as credible as the list of things it refuses to say. This slide is the list, and we would rather hand it to you than have you find it.

Not validated

One mechanism has survived one external falsifiable test. That is not a validated platform, and we never describe it as one. Our clinical-cohort outcome test came back null and stays on the record.

Not clinical

Research use only. No clinical validation exists, and nothing here predicts a patient's response to therapy. The decision layer is framed as clinical decision support and is not a cleared device.

Not experimentally confirmed

No prediction this system makes has been tested at a bench. The neoantigen calls, the binding predictions and the simulated immune dynamics are all in-silico today.

Not fully sourced

Parameter provenance is partial. Unsourced constants are flagged ESTIMATE rather than hidden, and several load-bearing numbers — immune kill rates among them — are still estimates.

Not GPU-resident, not agentic

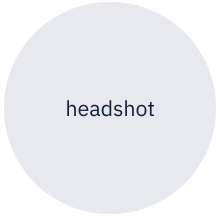
The agent-based engine runs on CPU; GPU paths are opt-in. The tumour board is deterministic with no ML model and no LLM. Both are accurate descriptions, not modesty.

No patent claim

We make none. Our contribution to the commons is the published Bio-USD schema under Apache 2.0, which is the part we would rather be judged on.

Every item above is recorded in the project's own engineering notes, which is why it can be stated this precisely. We would rather be the team that volunteers its falsification criteria than the one that gets caught lacking them.

Cognisom is built AI-first. That is how one founder covers genomics to simulation to rendering — and it is why the expertise we are adding next is scientific, not engineering.



David Walker

Founder and CEO, Eyentelligence, Inc. — AI-led development · Flower Mound, TX

- **Builds with AI as the primary method.** The whole pipeline — variant ingestion and somatic calling, HLA typing and neoantigen prediction, the ten-module agent-based tumour–immune engine, the literature-sourced parameter library, the OpenUSD and Omniverse layers — built by one founder working AI-first
- **With the discipline to match the speed:** 3,841 automated tests; optimisations verified bit-identical before shipping; scaling measured and re-measured rather than asserted; every parameter sourced to a PMID or flagged ESTIMATE
- Authored and published Bio-USD, the draft open OpenUSD schema for biological data, under Apache 2.0 with a citable DOI, engaged directly by Pixar's OpenUSD team
- **38 years in eyecare technology and supply chain** before this: Essilor of America 1989–2017; VP, AcuityLogic at Eyefinity 2019–2026, leading its first AI capabilities. Harvard Extension School ALM from September 2026, ALB expected 2027
- Cognisom is a **Harvard Innovation Labs** venture; Eyentelligence is an **NVIDIA Inception** member. **No formal scientific advisor is in place** — early conversations with a Harvard faculty member, nothing signed

What we are adding, and why

AI-first development moved the bottleneck. The engineering is ahead of the domain review, so **our open questions are now scientific rather than technical.** We are actively seeking research-level immunology, tumour-biology and translational collaborators to validate direction and tell us which mechanisms and which unmeasured constants deserve priority.

If selected

- Run the pre-registered peptide assay and publish the result either way
- Convene scientific reviewers on direction — the introductions matter as much as the credits
- Feed measured binding back into the library, replacing estimates with measurements

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